

Gabriel P. Oliveira

Data Scientist and Researcher

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Education

- 2021–2025 **Ph.D. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Research stay at **Politecnico di Torino, Italy** (2024–25)
Thesis: On the dynamics of music virality and its relation with mainstream success
Advisor: Prof. Mirella M. Moro, Ph.D. **GPA:** 94.9 out of 100
Co-advisor: Prof. Ana Paula Couto da Silva, Ph.D.
- 2019–2021 **M.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Thesis: Analyses of musical success based on time, genre and collaboration **GPA:** 92.0 out of 100
Advisor: Prof. Mirella M. Moro, Ph.D.
Co-advisor: Prof. Anisio Lacerda, Ph.D.
- 2015–2018 **B.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
GPA: 91.3 out of 100
- 2012–2014 **High School Technical Degree in Computer Networks**, *Centro Federal de Educação Tecnológica de Minas Gerais (CEFET-MG)*, Brazil.

Professional Experience

- 2025– **Tech Lead**, *Project SyntheticGIST*, UFMG/aequalis, Belo Horizonte, Brazil.
Lead the design and implementation of an AI-based platform for synthetic oncology data, leveraging probabilistic models, causal inference, and generative models to simulate patient trajectories. Developed and validated model pipelines, conducted performance evaluations, and coordinated experimentation cycles. Acted as the team's technical reference, aligning research, engineering outputs, and business requirements.
- 2024–2025 **Visiting Researcher**, *Politecnico di Torino*, Turin, Italy.
As a visitor at the SmartData@PoliTO center and the TNG Group, I developed epidemic-inspired modeling techniques for understanding virality patterns in large-scale social platforms. Applied statistical inference, algorithmic modeling, and time-series analysis to extract insights from unstructured behavioral data.
- 2021–2024 **Data Scientist**, *Analytical Capabilities Project*, UFMG/MPMG, Belo Horizonte, Brazil.
This project is a collaboration between the Computer Science Department of UFMG and the Prosecution Service of the State of Minas Gerais (MPMG). My activities included building ML and NLP-based models for fraud detection and public spending classification. Designed feature engineering processes, trained predictive algorithms, and evaluated model performance on complex, real-world datasets.
- 2019–2021 **Research Assistant**, *Project Både*, UFMG, Belo Horizonte, Brazil.
Applied network science, machine learning, and statistical analysis to model collaboration patterns and success dynamics in large-scale music industry datasets.
- 2019 **Data Science Intern**, *Data Science Residency Program*, UFMG/Localiza, Belo Horizonte, Brazil.
Developed ML models using clustering and association rule mining to support marketing and operational optimization in the car rental industry.
- 2018–2019 **Software Engineering Intern**, *dti digital*, Belo Horizonte, Brazil.
Developed data-driven applications by designing SQL databases and implementing RESTful APIs using C#.NET, with frontend integration in AngularJS.
- 2017–2018 **Undergraduate Research Assistant**, *Project FoRGit*, UFMG, Belo Horizonte, Brazil.
Built and analyzed heterogeneous collaboration graphs from GitHub data, designing graph-based metrics to quantify developer interaction patterns.
- 2016–2017 **Undergraduate Teaching Assistant**, *Computer Science Dept.*, UFMG, Belo Horizonte, Brazil.
Teaching assistant in the Computer Programming course (DCC001) for Engineering students.
- 2015 **IT Support Intern**, *Support Center for e-learning*, UFMG, Belo Horizonte, Brazil.
Network and server management, user support, and support to audiovisual content production.

Skills

AI	Generative AI (LLMs, Synthetic Data), Causal Inference, NLP
Data Science	Network Science, Statistical Modeling, Hypothesis Testing, Graph Mining, Time Series Analysis
Programming	Python, SQL, PySpark, C/C++, Git
Libraries	Scikit-learn, NetworkX, Pandas, NumPy, Matplotlib/Seaborn, PyTorch

Awards

- 2023 **Best Master Thesis Award**, Brazilian Symposium on Databases (SBBD).
Prize awarded in the 5th Contest of Theses and Dissertations on Databases.
- 2022 **Best Master Thesis Award – Finalist**, Congress of the Brazilian Computer Society (CSBC).
One of the ten Master theses selected for presentation at the Theses and Dissertations Contest of the Congress of the Brazilian Computer Society.
- 2018 **Best Graduating Student Award – 1st Runner-up**, Universidade Federal de Minas Gerais.
Prize awarded for students with the highest GPA in the Computer Science Undergraduate Program.

Selected Publications

The complete list of publications is available on my webpage.

- 2025 **Contagious rhythms: A wave-based epidemic approach for music virality on social platforms**, *Intl. Conf. on Advances in Social Networks Analysis and Mining (ASONAM)*, Niagra Falls, Canada.
GP Oliveira, L Vassio, APC da Silva, MM Moro
- 2025 **On the causal relationship between music virality and success**, *IEEE Access*.
GP Oliveira, APC Silva, MM Moro
- 2024 **What makes a viral song? Unraveling music virality factors**, *ACM Web Science Conference (WebSci)*, Stuttgart, Germany.
GP Oliveira, APC da Silva, MM Moro
- 2023 **PLUS: A Semi-automated Pipeline for Fraud Detection in Public Bids**, *Digital Government: Research and Practice (DGOV)*.
MA Brandão, APG Reis, BMA Mendes, CAB Almeida, **GP Oliveira**, H Hott, LD Gomide, LL Costa, MO Silva, A Lacerda, and GL Pappa
- 2023 **How do developers collaborate? Investigating GitHub heterogeneous networks**, *Software Quality Journal*.
GP Oliveira, AFC Moura, NA Batista, MA Brandão, A Hora, MM Moro
- 2020 **Detecting Collaboration Profiles in Success-based Music Genre Networks**, *International Society for Music Information Retrieval Conference (ISMIR)*, Montréal, Canada, **Best Paper Presentation Award**.
GP Oliveira, MO Silva, DB Seufitelli, A Lacerda, MM Moro

Languages

Portuguese	Native
English	Advanced
Italian	Conversational/Intermediate
Spanish	Basic

Other activities

- 2025 External Reviewer – BraSNAM, WEI
- 2024 External Reviewer – SEMISH, DSW
- 2023 External Reviewer – BraSNAM, SEMISH, DSW
- 2022 External Reviewer – BraSNAM, SBBD
- 2021 External Reviewer – WebMedia, SEMISH, WEI